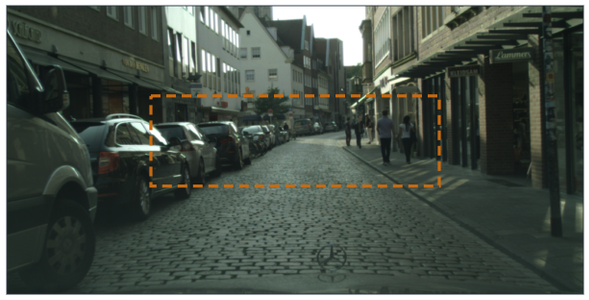
Stage outputs of F+AL on a Cityscapes test frame (munster_000150_000019)

Input frame



3 × 1024 × 2048
dashed: far band seen by the fovea view

How to read the panels

Feature maps: first three principal components of the channels, as RGB

People probability

Vehicle probability

Cell readouts: cells above 0.5, coloured by the more likely class, opacity = probability

Model: F+AL (runs/cityscapes_f_align/seed0), Cityscapes test (official val), ground truth: 39 People cells, 85 Vehicle cells

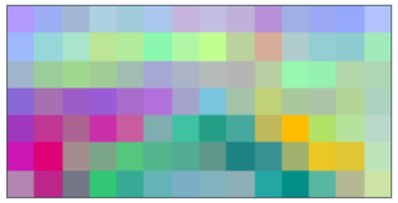
(i) Context view branch

Context view



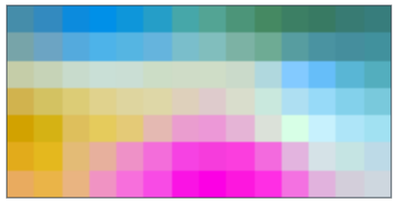
3 × 112 × 224

Backbone stride 16



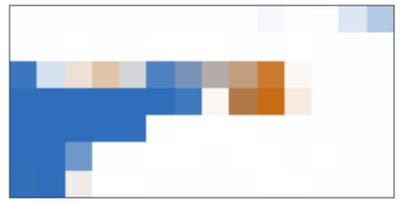
96 × 7 × 14
MobileNetV2 layers 0–13

Fused stride-16 map



1408 × 7 × 14
upsampled stride 32 + lateral

Logit maps z



People / Vehicle, 7 × 14
1×1 classifier, sigmoid

Cell readout r_v



all 256 cells
mask-weighted mean + bias

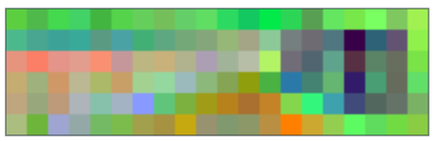
(ii) Fovea view branch

Fovea view



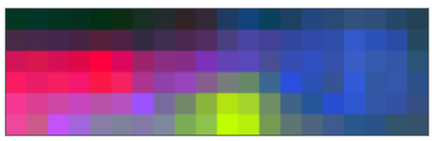
3 × 96 × 320

Backbone stride 16



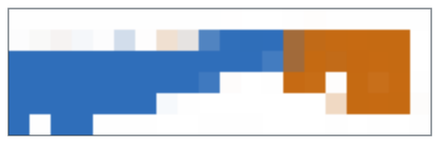
96 × 6 × 20
MobileNetV2 layers 0–13

Fused stride-16 map



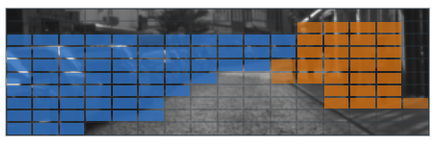
1408 × 6 × 20
upsampled stride 32 + lateral

Logit maps z



People / Vehicle, 6 × 20
1×1 classifier, sigmoid

Cell readout r_v



160 far cells
mask-weighted mean + bias

Σ

per-cell view blend
+ cell bias, sigmoid

Ground truth cells



from the full-resolution masks (≥ 1 px)

vs

Final cell output



256 cells × 4 · P+V F1 on this frame 91.2